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Use-Case Flow Specification

System: Internal Microservice Catalog & Health Portal | UC-02: Monitor Microservice Health

Use Case Name:	UC-02: Monitor Microservice Health
Primary Actors:	Automated Health Checker (System Actor), DevOps Engineer (Secondary Recipient)
Description:	Periodically pings registered health endpoints at 30s intervals, computes rolling 24h availability, and dispatches real-time downtime alerts upon 3 consecutive probe failures.

1. Preconditions

- Microservice is registered in catalog with valid health check endpoint URL.
- Automated health checker background scheduling worker is active and running.

2. Postconditions

- Service status (Healthy / Unhealthy) and latency logs updated in database.
- Rolling 24-hour service availability metrics re-calculated.
- On 3 consecutive failures, an active incident is logged and alert dispatched to DevOps.

3. Main Success Scenario (Step-by-Step)

Step 1 (Trigger):	Automated Health Checker triggers periodic check at 30-second interval.
Step 2 (Fetch):	System retrieves list of active microservices and registered health check endpoints.
Step 3 (Probe):	System sends HTTP GET request to target microservice health endpoint.
Step 4 (Response):	Microservice responds with HTTP 200 OK and valid JSON health payload within 3s.
Step 5 (Status Update):	System marks microservice status as 'Healthy' and logs probe latency.
Step 6 (Metrics):	System recalculates rolling 24-hour service availability percentage.
Step 7 (Dashboard):	System updates live monitoring portal UI accessible to DevOps Engineers.
Step 8 (Completion):	Use case completes successfully for the current 30-second cycle.

4. Alternate Flows

4a. Health Probe Failure / Downtime Detected

4a1. In Step 4, microservice probe times out (>3s) or returns non-200 HTTP status (500/503).

4a2. System increments consecutive failure counter for the microservice.

4a3. If failures < 3, system logs Warning state and waits for next 30-second cycle.

4a4. If failures = 3, system updates status to 'Unhealthy' and logs active incident.

4a5. System triggers <<extend>> UC-05: Send Downtime Alerts via Slack webhook and email.

4a6. Monitoring continues every 30s until HTTP 200 OK is received on consecutive checks.

4a7. Upon recovery, system marks status as 'Healthy', records downtime duration, dispatches recovery notice, and resumes at Step 6.

3a. Invalid or Unreachable Health Endpoint

3a1. In Step 3, probe fails due to DNS resolution error or invalid URL structure.

3a2. System flags catalog entry with 'Configuration Error' and sends alert to registering DevOps Engineer.